

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA07

COURSE OUTCOME COVERED

CO4:

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards
(BL2,BL6)

Assessment Tool Weightage: 30%

QUESTIONS: 5 Total Marks:25 Annexure A

Comparative Analysis

1. Star topology is generally more expensive than bus topology because it requires a central device such as a switch or hub and more cabling to connect each node individually. In contrast, bus topology uses a single backbone cable, making it low-cost and easier to install. However, in terms of reliability, star topology is superior because failure of one node does not affect others, whereas a failure in the main bus cable can bring down the entire network. Performance is also better in star topology since data transmission is managed through a central device, reducing collisions, while bus topology suffers from performance degradation as more devices are added due to increased data traffic on the shared medium.
2. Mesh topology offers very high fault tolerance because every node is connected to multiple other nodes, providing multiple paths for data transmission. If one link fails, data can still be routed through alternate paths without disrupting communication. In contrast, star topology has moderate fault tolerance; while failure of individual nodes does not affect the network, failure of the central hub or switch can cause the entire network to stop functioning. Therefore, mesh topology is more robust and reliable in critical environments compared to star topology.
3. Bus topology is simple and easy to install because it uses a single backbone cable with minimal connections, making it suitable for small networks. On the other hand, mesh topology is highly complex to install as each node must be connected to every other node, requiring a large number of cables and ports. This complexity increases significantly as the number of devices grows, making mesh topology time-consuming and costly to implement compared to the straightforward setup of bus topology.
4. Hybrid topology is highly scalable because it combines two or more different topologies (such as star, mesh, or bus), allowing the network to expand flexibly according to organizational needs. New segments can be added without affecting the existing network structure. In comparison, star topology is also scalable but to a limited extent, as it depends on the capacity of the central device (switch or hub). Once the central device reaches its port limit, expansion requires additional hardware. Thus, hybrid topology provides greater scalability than star topology.
5. In terms of maintenance, star topology is relatively easy to manage because issues can be quickly identified and isolated to a specific node or cable. Mesh topology, while highly reliable, is difficult to maintain due to its complex structure and large number of connections, making troubleshooting time-consuming. Bus topology is harder to maintain than star because faults in the backbone cable can disrupt the entire network and are difficult to locate. Hybrid topology has moderate maintenance complexity; while it

benefits from flexibility, managing multiple combined topologies requires careful monitoring and skilled administration.

RUBRICS FOR CALCULATION

Criteria		Weightage Expected	Proficient (3) D		
Understanding of Concepts	15%	Demonstrates complete and in depth understanding of all concepts being compared	Good understanding with minor gaps	Basic understanding; some concepts unclear	Poor understanding of concepts
Comparison Depth	20%	Provides detailed, insightful, and meaningful comparisons across multiple aspects	Adequate comparison with relevant points	Limited comparison; lacks depth	Minimal or incorrect comparison
Criteria Selection	10%	Selects highly relevant and appropriate comparison parameters	Mostly relevant criteria	Some irrelevant or missing criteria	Poor or no criteria selection
Analytical Skills	15%	Strong analysis with logical reasoning and clear justification	Good analysis with some justification	Basic analysis with weak reasoning	No clear analysis
Organization & Structure	10%	Well-organized, clear, and logically structured	Generally organized	Somewhat disorganized	Poorly structured
Use of Examples / Evidence	10%	Uses strong, relevant examples or data to support comparison	Uses some examples	Limited examples	No supporting examples
Clarity of Presentation	10%	Very clear, concise, and easy to understand	Mostly clear	Somewhat unclear	Difficult to understand
Conclusion & Insights	10%	Insightful conclusion with strong	Clear conclusion	Basic conclusion	No or weak conclusio

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Question 1 – Star Topology vs Bus Topology

Compare Star topology and Bus topology in terms of cost, reliability, and performance.

Answer

Cost

Star topology is more expensive because every device requires a separate cable connected to a central switch or hub. Bus topology is cheaper because all devices share a single backbone cable.

Reliability

Star topology is more reliable because failure of one computer or its cable does not affect the remaining devices. In bus topology, failure of the main backbone cable can stop communication across the entire network.

Performance

Star topology provides better performance because the switch manages communication between devices and reduces collisions. Bus topology uses a shared communication medium, so performance decreases as more devices and traffic are added.

Criterion	Star Topology	Bus Topology
Cost	Higher; requires a central device and individual cables.	Lower; uses a shared backbone cable.
Reliability	Higher; individual node failures are isolated.	Lower; backbone failure can affect the entire network.
Performance	Better; communication is managed by a central switch.	Lower as traffic and devices increase.

Conclusion

Star topology is better for modern networks because it provides higher reliability and performance, although it has a higher implementation cost.

Question 2 – Mesh Topology vs Star Topology

Compare Mesh topology and Star topology in terms of fault tolerance and reliability.

Answer

Mesh topology provides very high fault tolerance because devices have multiple connections and alternative paths are available for data transmission. If one link fails, traffic can be redirected through another available path.

Star topology provides moderate fault tolerance. Failure of an individual computer or cable affects only that device, but failure of the central switch or hub can affect the entire network.

Feature	Mesh	Star
Fault tolerance	Very high	Moderate
Alternative paths	Multiple	Usually single
Central dependency	Low	High
Reliability	Very high	High
Cost	Very high	Moderate

Conclusion

Mesh topology is more reliable than star topology for critical environments because it provides multiple alternative communication paths.

Question 3 – Bus Topology vs Mesh Topology

Compare Bus topology and Mesh topology in terms of installation complexity.

Answer

Bus topology is simple and easy to install because all devices share a single backbone cable. It requires fewer cables and connections.

Mesh topology is much more complex because every device must have connections to multiple or all other devices. As the number of devices increases, the number of required connections increases rapidly.

Full-Mesh Link Calculation

Number of links = $n(n - 1) / 2$

For 5 devices: $5(5 - 1) / 2 = 10$ links.

Feature	Bus	Mesh
Installation	Simple	Complex
Cabling	Low	Very high
Cost	Low	Very high
Scalability	Limited	Difficult
Maintenance	Relatively simple	Complex

Conclusion

Bus topology is easier and cheaper to install, while mesh topology provides better redundancy at the cost of greater complexity and expense.

Question 4 – Hybrid Topology vs Star Topology

Compare Hybrid topology and Star topology in terms of scalability.

Answer

Hybrid topology combines two or more topologies such as star, mesh, or bus. This allows organizations to expand different sections of the network according to their requirements. New network segments can be added without completely changing the existing infrastructure.

Star topology is also scalable, but its expansion depends on the available ports and capacity of the central switch. When all ports are occupied, another switch must be added.

Feature	Hybrid	Star
Scalability	Very high	High
Expansion	Flexible	Depends on switch capacity
Flexibility	Very high	Moderate
Additional hardware	Based on design	Required when ports are exhausted
Suitable for large organizations	Excellent	Good

Conclusion

Hybrid topology provides greater scalability and flexibility than star topology, making it more suitable for large and complex organizations.

Question 5 – Maintenance Comparison

Compare Star, Mesh, Bus, and Hybrid topologies in terms of maintenance.

Answer

Star Topology

It is relatively easy to maintain because each device has an individual connection to the central switch. A faulty device or cable can usually be identified quickly.

Mesh Topology

It is difficult to maintain because there are many connections between devices. Identifying and troubleshooting a failed link can take more time.

Bus Topology

It is more difficult to maintain because a problem with the backbone cable can affect the entire network. Locating the exact fault can also be challenging.

Hybrid Topology

It has moderate maintenance complexity because it combines different topology types. Administrators need to understand and monitor each section of the network.

Topology	Maintenance	Main Reason
Star	Easy	Individual connections simplify fault isolation.
Mesh	Difficult	Large number of interconnected links.
Bus	Difficult	Backbone failure affects multiple devices.
Hybrid	Moderate	Combines multiple topology structures.

Conclusion

Star topology is the easiest to maintain, mesh is the most complex, bus can be difficult because of backbone faults, and hybrid requires skilled administration because multiple topology types are combined.